

Daniel Lofeodo

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Experience

2025-04 – Machine Learning Engineer (Fixed-Term Contract)

2026-05 *Hydro-Québec, Montreal, Canada*

RL and forecasting agents for hydro-dam releases under extreme, out-of-distribution (1-in-10,000-year) flood conditions.

- Built an end-to-end forecasting pipeline (XGBoost → transformer/Chronos); found supervised baselines failed to generalize out-of-distribution.
- Developed a constrained RL framework (PPO, dynamic reward shaping) that kept water levels within safety bounds in 9 of 9 extreme scenarios, vs. 2 of 9 for SL baselines.
- Outperformed the legacy deterministic tool's max-outflow assumption: 83% fewer unnecessary releases at equivalent or better safety margins.

2021-06 – Software Engineer

2024-08 *Creaform, Montreal, Canada*

Ultra precision 3D scanners and software for engineering.

- Developed core features for aerospace-grade 3D inspection software, including geometric tolerancing (circularity, cylindricity, flatness) for validating scanned parts against engineering specifications.
- Built a new industrial pipe inspection product by repurposing internal components, contributing to a high-revenue product line.
- Led UI/UX development for Creaform.OS, improving performance through profiling and delivering customizable interfaces.
- Contributed to large-scale C++ codebases in an Agile environment with CI/CD, code reviews, and continuous refactoring.

Education

2024-08 – M.Sc. Computer Science (Machine Learning)

2026-05 *Montreal Institute for Learning Algorithms, University of Montreal, Montreal, Canada*

- 4.1/4.3 GPA

2017-08 – B.Eng. Mechanical Engineering

2021-05 *Concordia University, Montreal, Canada*

- Dean's List (2021)

Spotlight Projects

2026-04 – Perspective Blind – 3D Spatial Reasoning Benchmark for VLMs

2026-05 Benchmark testing VLM spatial reasoning from an imagined, unseen viewpoint using 3D scene reconstructions.

- Built a benchmark testing viewpoint-invariant spatial reasoning across 5 VLM families on ScanNet 3D reconstructions.
- Found lateral-relation accuracy collapses from 83.6% to 8.1% – below random chance – as viewpoint misalignment increases.
- Built eval infra: schema-enforced outputs, consistency/confusion-matrix metrics, cost/latency tracking across providers.

2026-05 – Latent SpaceMail – Autonomous Multi-Agent Newsletter Pipeline

2026-07 Agentic pipeline delivering a personalized weekly AI news and research briefing; live 2 months.

- Orchestrated 6 Claude agents on GCP (Cloud Run, Pub/Sub, Firestore) in a fan-out/fan-in pipeline across parallel news/arXiv chains.
- Engineered tool-calling, concurrency control, and retry/backoff to score ~800 articles + 250 papers weekly, sustaining a 70%+ open rate.
- Built a Google-auth subscriber site with bilingual, topic-tiered preferences.

Technical Skills

Programming Languages:

Python, C++, C#, SQL, Java, JavaScript, TypeScript

AI / ML:

PyTorch, AutoML, scikit-learn, xarray, XGBoost, Pandas, Numpy, stable-baselines3, gymnasium, generative AI, machine learning, neural networks

Data science / Mathematics:

Statistical analysis, EDA, ETL, algorithms, linear algebra

Visualization Frameworks:

matplotlib, Seaborn, Plotly

Web/App Frameworks:

Streamlit, Flask, FastAPI, React

MLOps:

Azure, GCP, AWS, GitHub, Databricks, Docker, Weights & Biases, Tensorboard

Environments:

Windows, WSL, Linux

Project management:

Asana, Jira, Teams

Certifications

- AWS MLA (Cloud-native ML)
- Claude Code in Action
- Complete Modern C++ (C++11/14/17)

Conferences

- [Autonomous Hydroelectric Dam Operation under Extreme Flooding](#), Mila Annual Conference

Languages

- English (Native)
- French (Native)
- Italian (Advanced - C1)